

Differential Expression Analysis of the transcriptome of human Trabecular meshwork cells stimulated with Transforming Growth Factor β -1 (TGF- β 1) involved in Pseudoexfoliation Syndrome (XFS) and Glaucoma (XFG)

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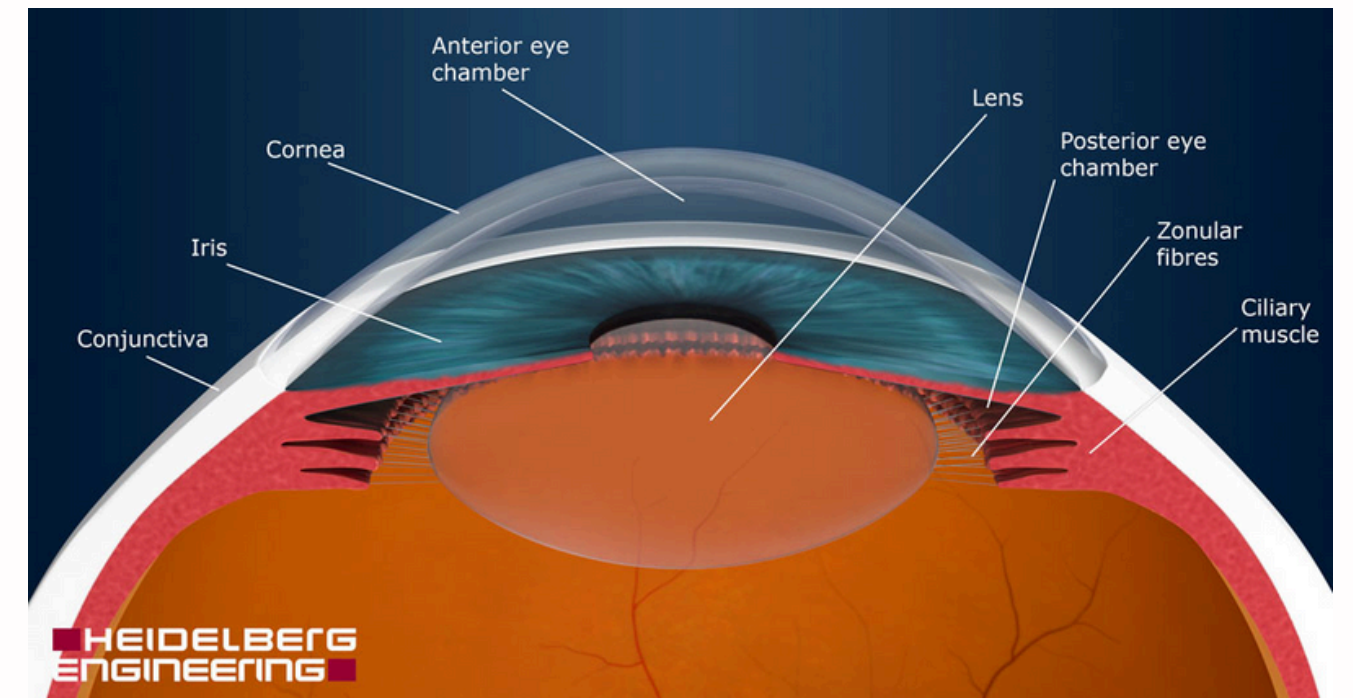
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1. INTRODUCTION

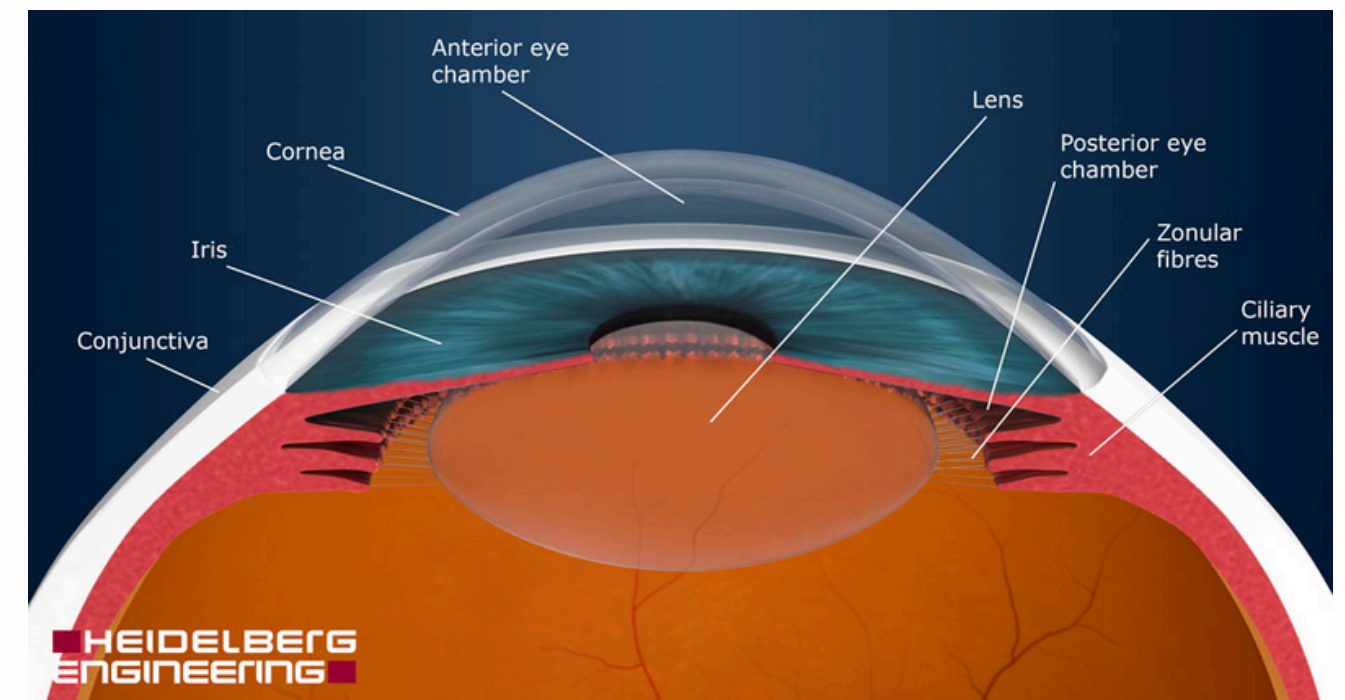
- **Pseudoexfoliation syndrome (XFS)** → Pseudoexfoliation Glaucoma (XFG)



Heidelberg Engineering. (n.d.). Anterior part of the eye [Photograph].
<https://www.know-the-eye.com/the-eye/anterior-part-of-the-eye/>

1. INTRODUCTION

- **Pseudoexfoliation syndrome** (XFS) → Pseudoexfoliation Glaucoma (XFG)
 - “Genome-Wide RNA Sequencing of Human Trabecular Meshwork Cells Treated with $TGF-\beta 1$: Relevance to Pseudoexfoliation Glaucoma” study by Roodnat AW et.al
 - Samples:
 - 4 Treated with $TGF-\beta 1$
 - 4 Controls, untreated
- } Paired data

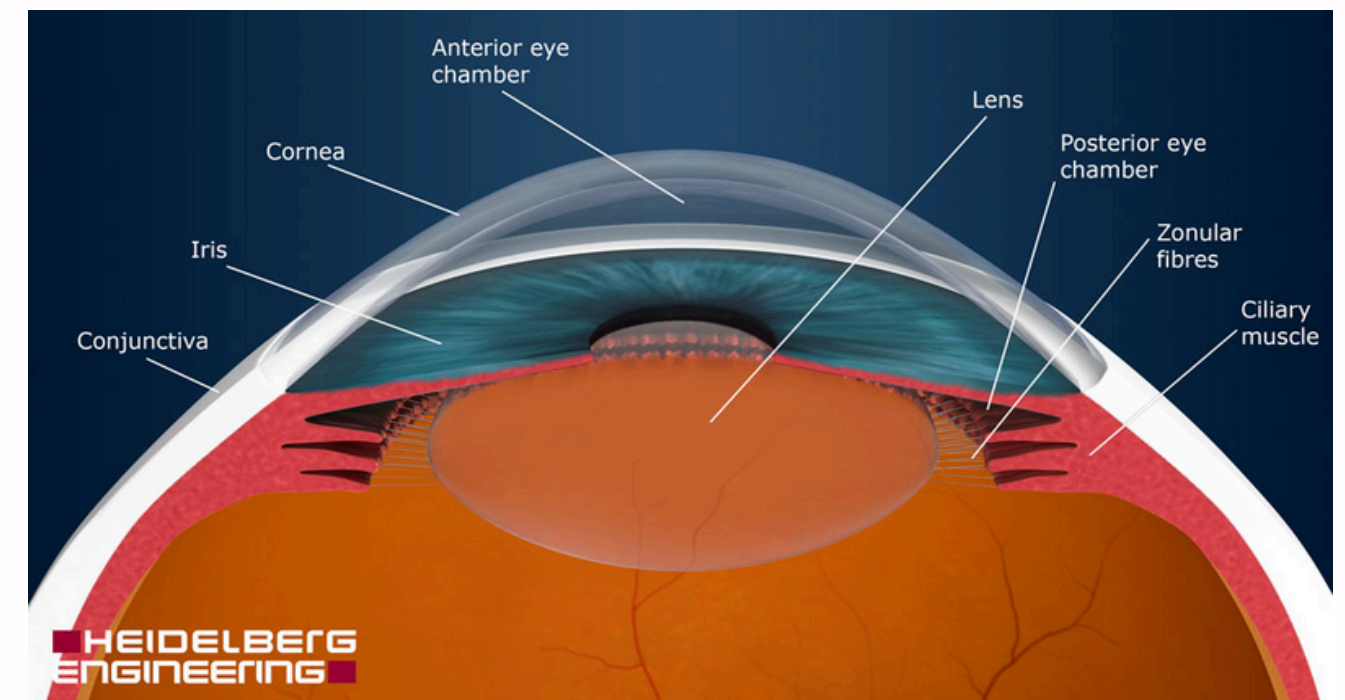


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1. INTRODUCTION

- **Pseudoexfoliation syndrome** (XFS) → Pseudoexfoliation Glaucoma (XFG)
- “Genome-Wide RNA Sequencing of Human Trabecular Meshwork Cells Treated with $TGF-\beta 1$: Relevance to Pseudoexfoliation Glaucoma” study by Roodnat AW et.al
- Samples:
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Aim → Understand which genes and biological processes are activated or inhibited in XFG.



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2. MATERIALS & METHODS

BLOCK 1: GENE AND PROTEIN CHARACTERISATION



Biomedical databases



Compare sequences



Information about
proteins



Genomic annotations

2. MATERIALS & METHODS

BLOCK 2: RNA-SEQ ANALYSIS WITH GALAXY



Obtention SRA RNA-seq



FastQC and MultiQC



Trimming and MultiQC



Alignment to GRCh38



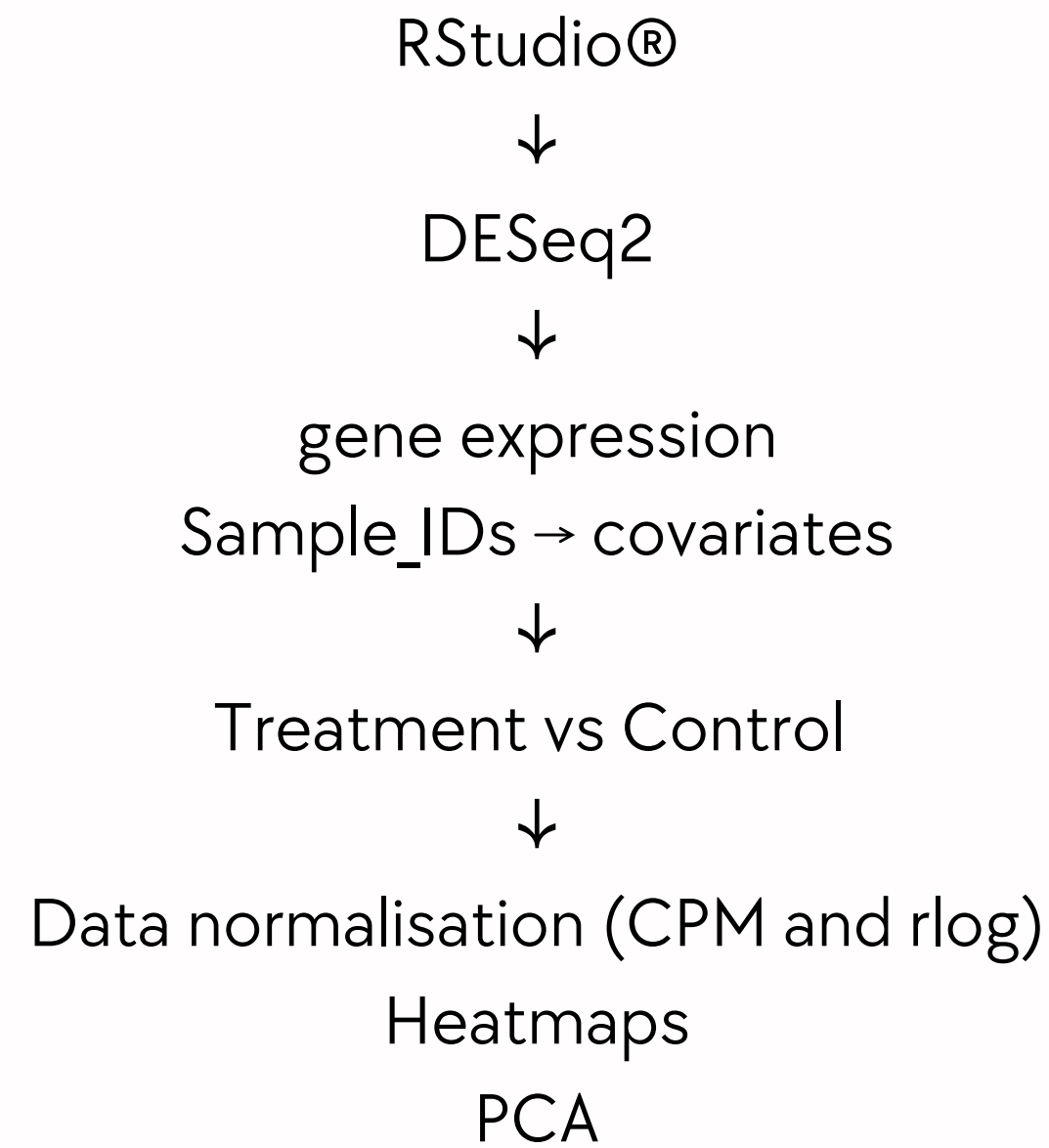
Gene quantification



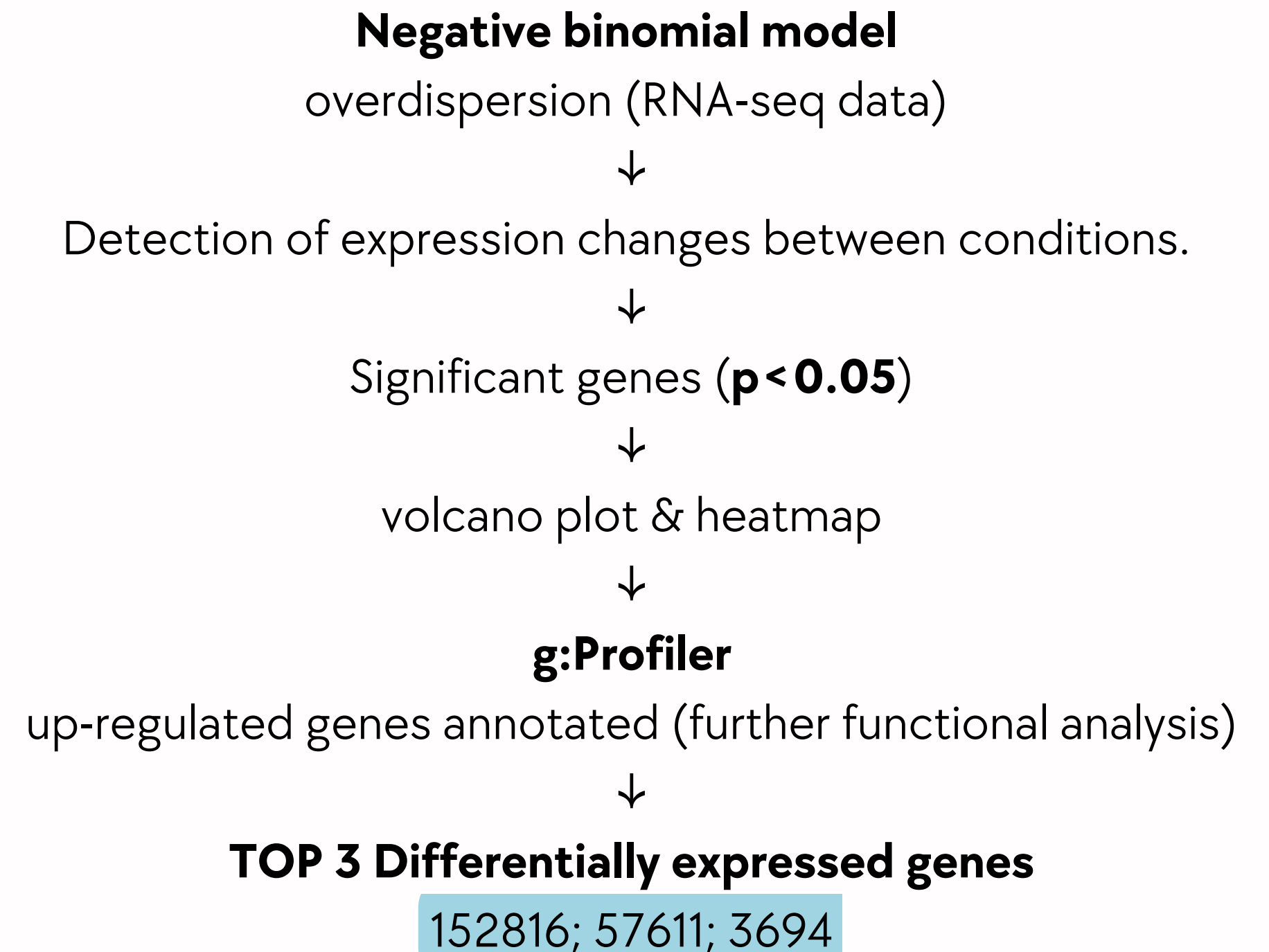
featureCounts

BLOCK 2: RNA-SEQ ANALYSIS WITH R

1. Pre-processing and global understanding of the data



2. Differential expression analysis with DESeq2



BLOCK 2: RNA-SEQ ANALYSIS WITH R

3.GO enrichment

up- and down-regulated genes

topGO

enriched GO terms



log2FoldChange thresholds

FDR correction



Genes associated with the most relevant GO terms



annotated & intersected

highlight key functional categories



Potential correlations with TGF- β 1

Top 3 up-regulated genes.

2. MATERIALS & METHODS

BLOCK 3: STRING

Protein **interactions** analysis by mapping associations between proteins



3. RESULTS & DISCUSSION

Block 1

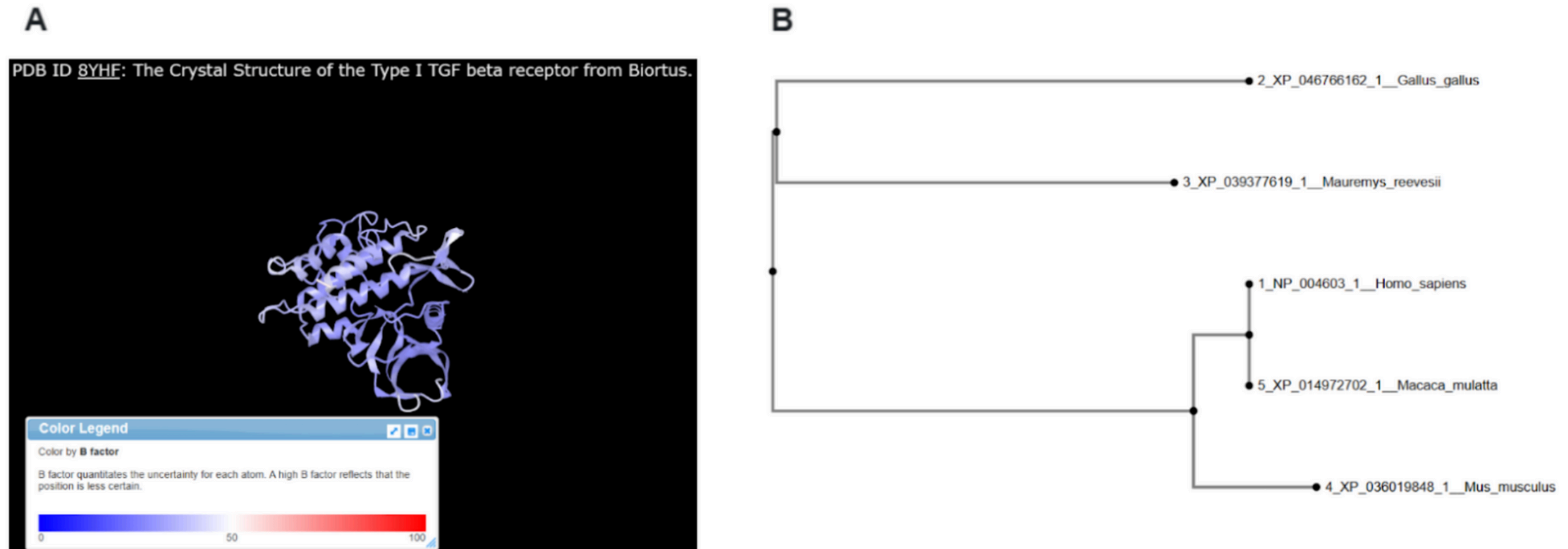


Fig. 1 Protein characterisation. (A) The crystal structure of the Biortus TGF beta type I receptor, stained for B factor, shows predominantly blue, indicating low B factors, suggesting that the protein structure is stable and reliable. (B) The phylogenetic tree shows the distance between species, showing the poricity between Macaca mulatta, and Homo sapiens. Gallus gallus and Mauremys reevesii could be considered ancestral

3. RESULTS & DISCUSSION

Block 2

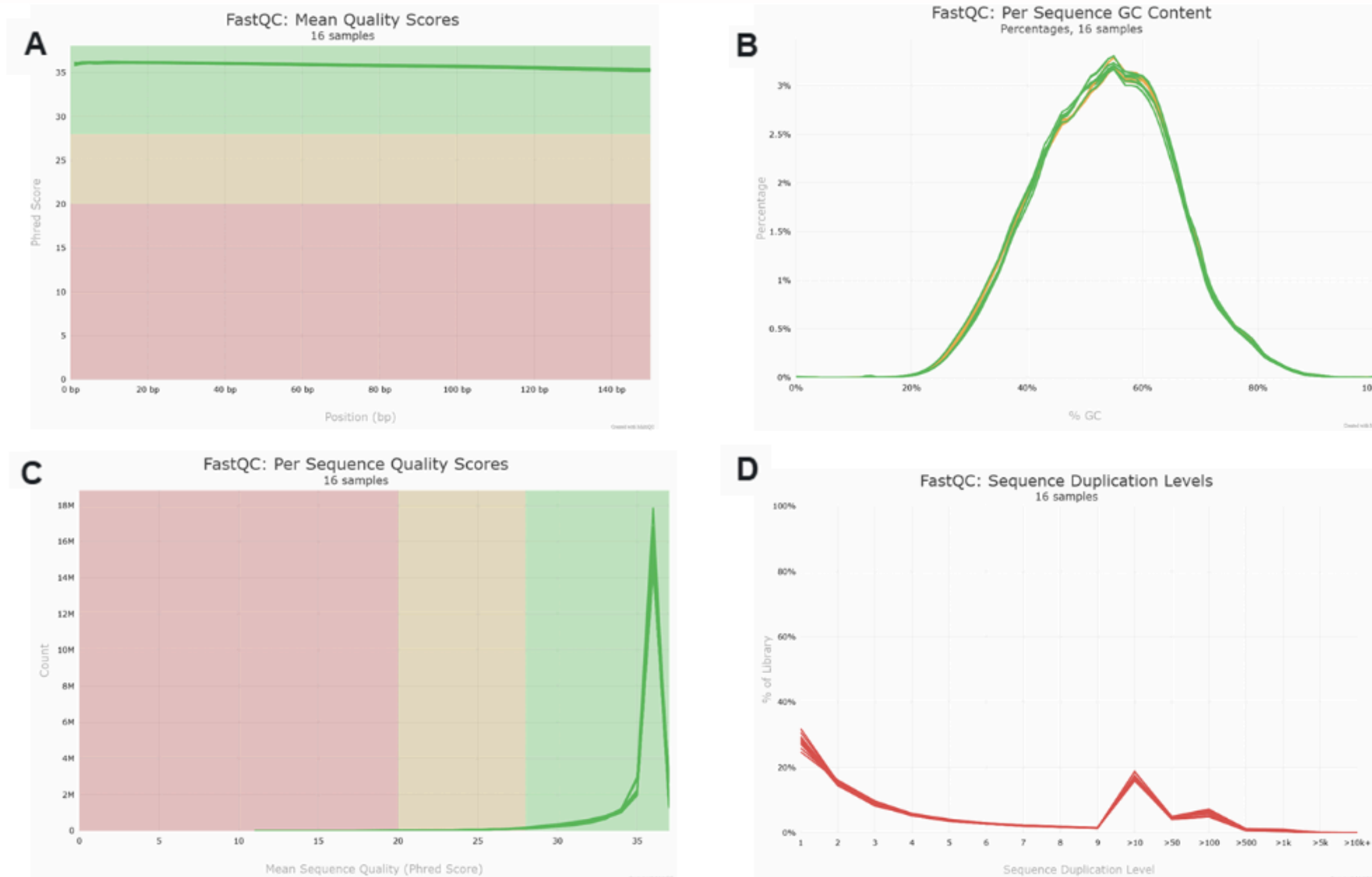


Fig. 2: Evaluation of the quality of sequencing reads for 16 samples using FastQC. (A) Average quality per position (Phred score) across reads, showing high and stable quality (>30) at all positions. (B) Distribution of GC content per sequence, with a normal curve centred around 50–55%, indicating the absence of contamination or biases in nucleotide composition. (C) Average quality per read, with most reads presenting values higher than 35, reflecting high overall quality. (D) Levels of sequence duplication, with a relevant proportion of duplicate reads, possibly due to high expression of specific transcripts or low library complexity. (A) & (C) Red, yellow and green areas represent low, medium and high quality, respectively. Created with Galaxy European Server

3. RESULTS & DISCUSSION

Block 2

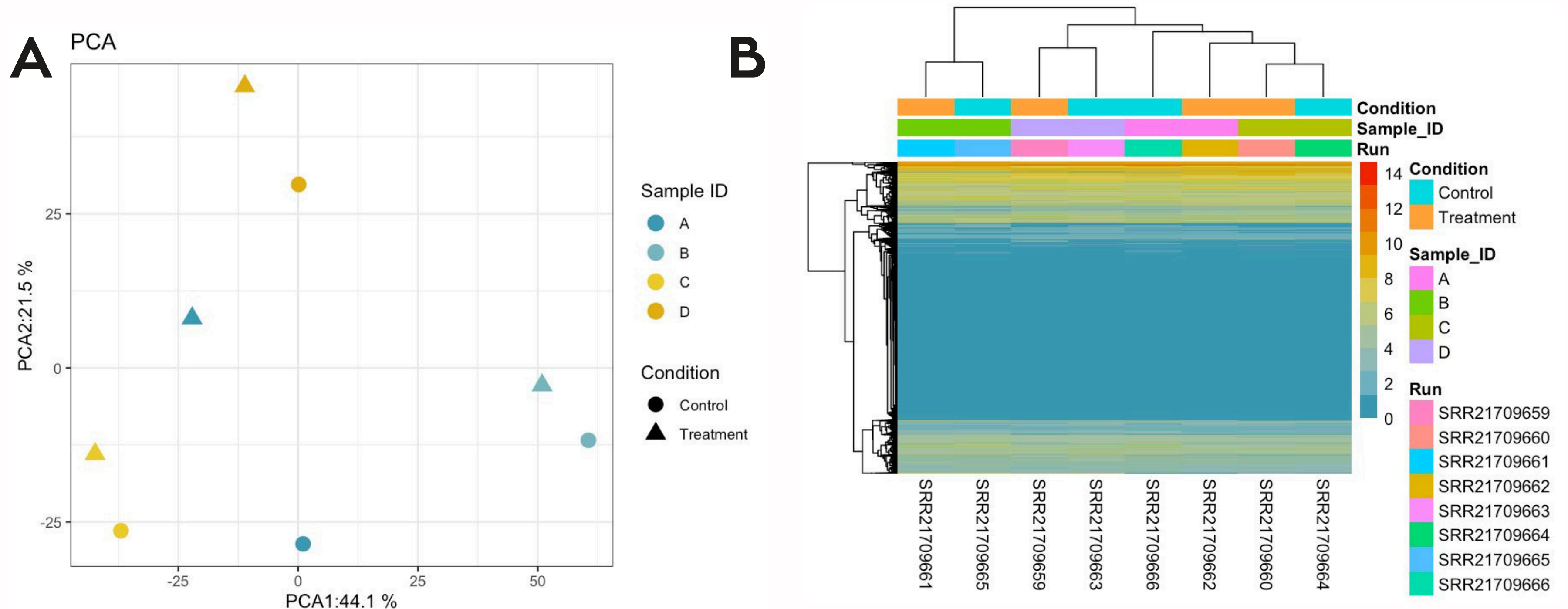


Fig. 3: Principal component analysis (PCA) and HeatMap of normalised transcriptome data. (A) Samples are represented according to experimental condition (by shape) and corresponding patient (by colour). The x and y axes show the first two components, which have 44.1% and 21.5% of the total variance, respectively. The clustering of the samples indicates a clear separation between conditions, with some variability between patients. (B) This plot shows a gene expression profile from RNA sequencing data normalised by applying Counts Per Million (CPM). Each column represents a sample (Run ID), and each row corresponds to a gene. The colour scale indicates the expression level, with red meaning high expression and blue representing low expression. The annotations above the samples provide information from the metadata, including Condition (Control and Treatment), Sample_ID (A, B, C and D), and Run (ID of the samples). Created with RStudio®

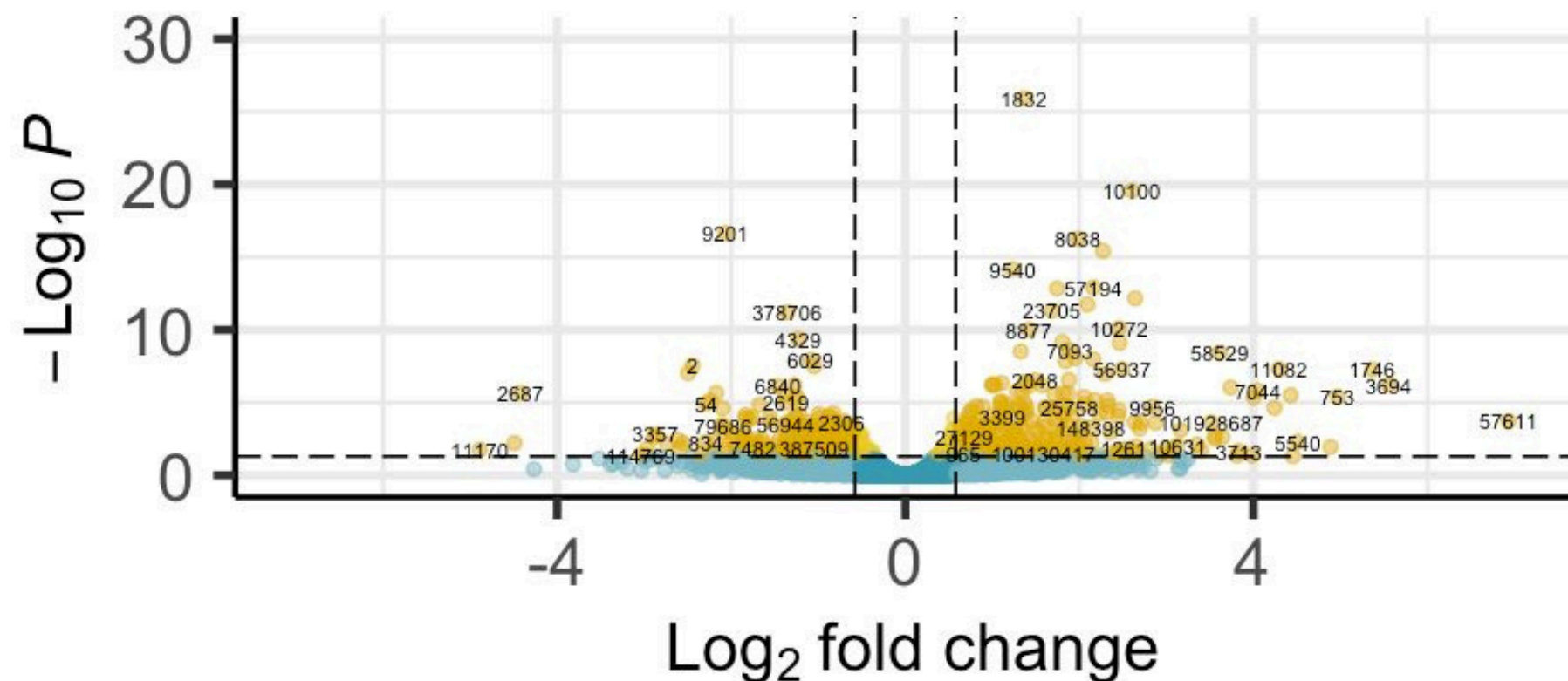
3. RESULTS & DISCUSSION

Block 2

Volcano Plot Control vs Treatment

EnhancedVolcano

● NS ● Log₂ FC ● p-value ● p – value and log₂ FC



The most significant genes regarding DE

- 152,816 → **ODAPH** (odontogenesis-associated phosphoprotein)
- 57,611 → **ISLR2** (immunoglobulin superfamily containing leucine-rich repeat 2)
- 3,694 → **ITGB6** (integrin subunit beta 6)

Fig 4: Volcano Plot representing Control group vs Treatment group genes. The X-axis represents the Log2FoldChange, and it represents the magnitude of gene expression differences; the values further to the right side are more expressed than those on the left side of the graph. The Y-axis represents the statistical significance of differential expression; the higher the value, the more significant the differential gene expression.

3. RESULTS & DISCUSSION

Block 2

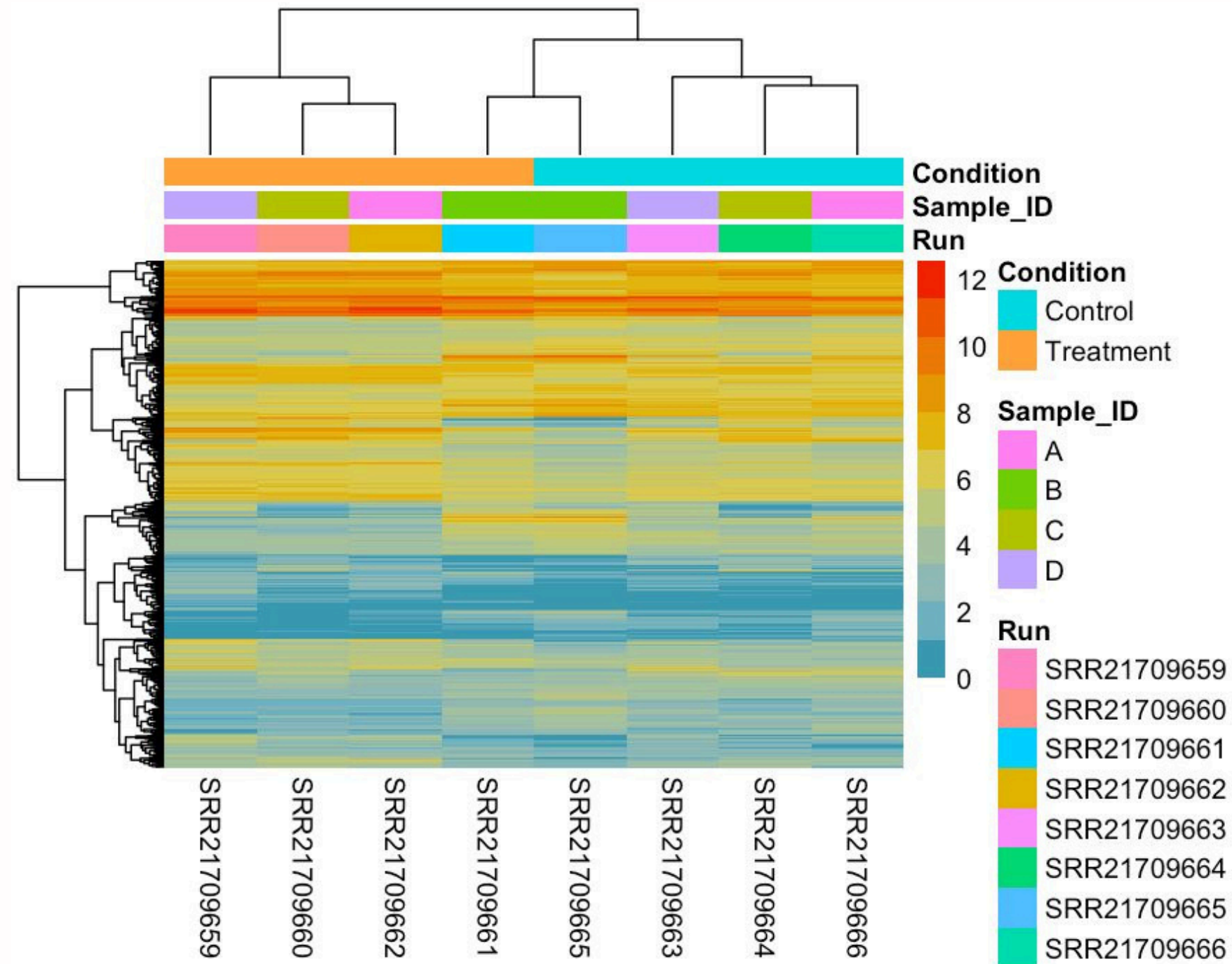


Figure 5: HeatMap of differentially expressed genes. This plot shows a gene expression profile from differentially expressed genes, normalised by applying Counts Per Million (CPM). Each column represents a sample (Run ID), and each row corresponds to a gene. The colour scale indicates the expression level, with red meaning high expression and blue representing low expression. The annotations above the samples provide information from the metadata, including Condition (Control and Treatment), Sample_ID (A, B, C and D), and Run (ID of the samples). Created with Rstudio®

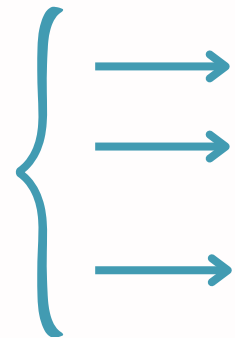
3. RESULTS & DISCUSSION

Block 2

Table 1, Table 2: The first column (GO.ID) is the code of the term in the GenOntology database. The term is the main function of the genes found within. The annotated column indicates the total number of genes from the input list associated with the GO term. The fourth column presents how many of the annotated genes the program has found to be statistically significant. The Expected column indicates the number of genes expected to be included in that term by chance. The last column presents a p-value, if it is low, it indicates that the genes do belong to the category.

Table 1: GO.ID enrichment results table for DE gens

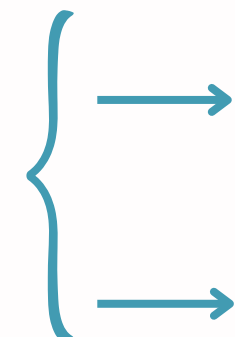
ODAPH
ISLR2
ITGB6



	GO.ID	Term	Annotated	Significant	Expected	classicFisher
1	GO:0032501	Multicellular organismal process	4829	390	272.07	2.1e-19
2	GO:0048856	Anatomical structure development	4306	357	245.28	5.9e-19
3	GO:0048731	System development	3016	272	171.8	5.2e18
4	GO:0007275	Multicellular organism development	3508	300	199.82	7.9e-17
5	GO:0009888	Tissue development	1425	156	81.17	1.0e-16

Table 2: GO.ID enrichment for up-regulated genes in the Treatment samples results table by log2FoldChange >+2.

ITGB6



	GO.ID	Term	Annotated	Significant	Expected	classicFisher
1	GO:0007178	Cell surface protein Ser-Thr	7	7	4.03	0.0018
2	GO:0141091	TGF- β receptor	7	7	4.03	0.018
3	GO:0090092	Regulation of transmembrane receptor protein	6	6	3.545	0.033
4	GO:0006022	Aminoglycan metabolic process	5	5	2.88	0.059
5	GO:0007179	TGF- β receptor	5	5	2.88	0.059

3. RESULTS & DISCUSSION

Block 3

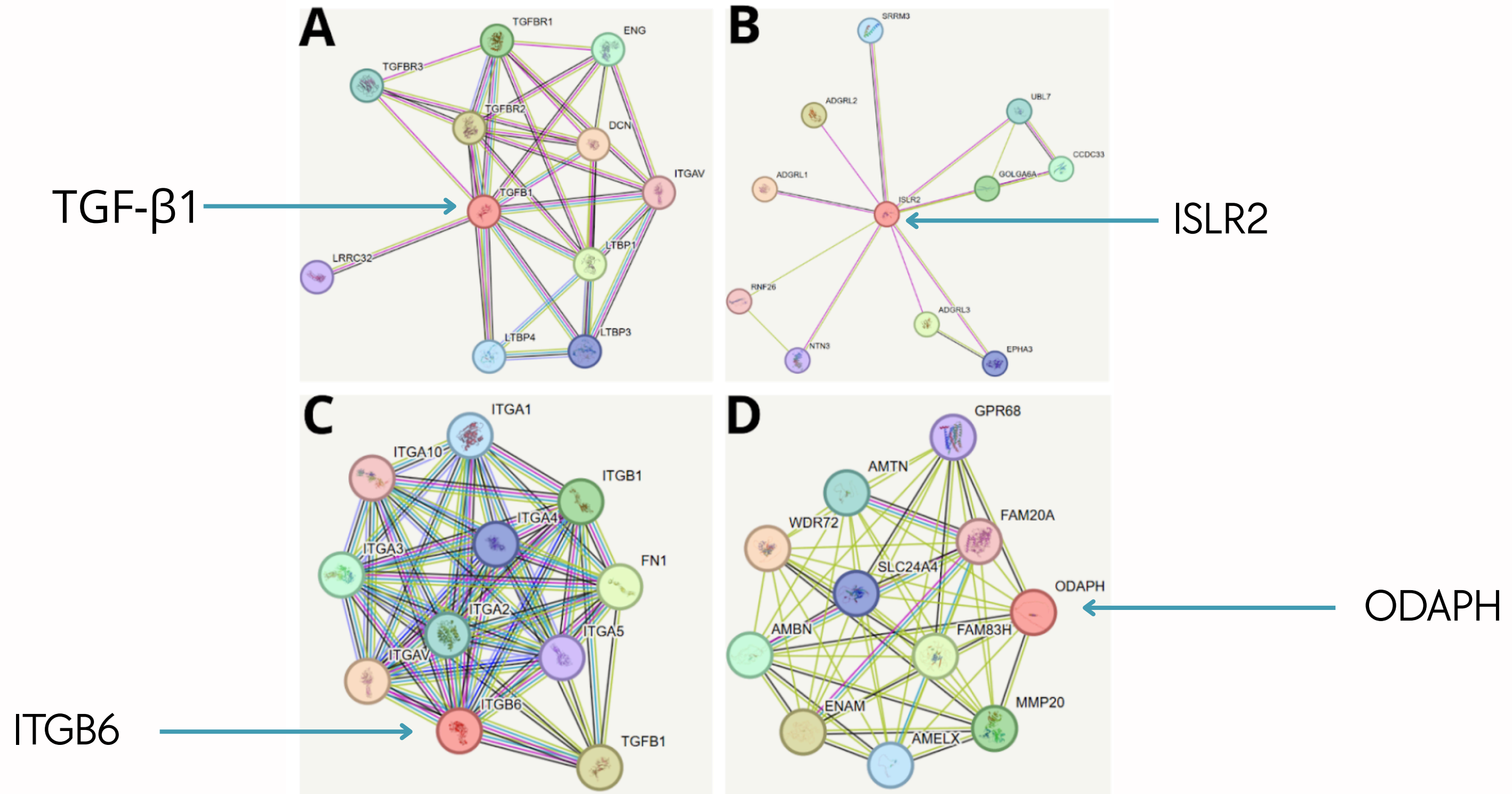


Fig. 5. TGF- β 1, ISLR2, ITGB6, and ODAPH association networks generated using STRING. Nodes represent proteins and are colored according to their interaction stage or functional context. Edges indicate associations based on various evidence channels in STRING; these represent functional interactions and do not necessarily imply direct physical binding. (A) TGF- β 1 network, (B) ISLR2 network, (C) ITGB6 network, (D) ODAPH network.

4. CONCLUSIONS

- Aim achieved → identify key genes & pathways
- **ITGB6**
 - associated with **XFG** → role in ECM remodelling
 - interaction with **TGF- β 1** → shown in the STRING network analysis.

4. CONCLUSIONS

Limitations

1. Reduced number of samples
2. Sex and Age not researched
3. Limited experience with RStudio and online tools

Future perspectives

1. Increase the number of samples
2. Study ITGB6's role

Thank you for your attention!

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